Take-home: Evaluate the Grader

VinylEquity · Evaluation Engineer role · September 2026

> **Time cap: 4 hours.** We'd rather see 3 sharp hours than 8 polished ones — stop at the cap and note what you'd do with more time.

Background

Billfold (a fictional B2B invoicing platform) uses an AI grader in its CI: for each pull request, the grader reads a ticket's requirement, inspects the code, and issues a verdict — **PASS**, **FAIL**, or **OUT_OF_SCOPE** — along with the sentence from the requirement it relied on (`ai_quote`) and a one-line rationale. A FAIL blocks the merge until a human intervenes.

`verdicts.json` (provided alongside this document) contains 60 recent verdicts. For each, a human adjudicator later recorded the final label (`human_label`) and a note. Treat the human labels as ground truth — with one caveat you'll discover in the data.

Your tasks

1. **Measure the grader.** Choose and compute the metrics you would actually report to this team. Explain briefly why you chose them — and, if you rejected an obvious metric, why.
2. **Find the failure pattern(s).** Not "the grader is sometimes wrong" — name each systematic failure class you find, list the record IDs in it, and describe the mechanism.
3. **Build one automated check** (any language, small is fine) that would catch one of your failure classes *before* a human ever sees the verdict. Include how many of the 60 records it flags, and its false positives if any.
4. **Write a one-page report** for the team: what you found, what you recommend, what you deliberately did *not* measure or conclude from this dataset, and what data you'd want next.

Rules

- **AI tools are allowed and encouraged.** But include a short section: *where your AI assistant was wrong or misleading while you worked, and how you caught it*. This role is about evaluating AI output — we mean it.
- Cite record IDs for every claim. A pattern without IDs is an impression, not a finding.

- Deliverable: a folder or repo with your code, your report (markdown is fine), and a README saying how to run the check.

Submission & confidentiality

- Submit a zip (or a private repo link) **by replying to the email this assignment arrived from**, within **5 days** of receiving it. If you need a different window, just ask — flexibility is fine, silence isn't.
- Please keep the assignment and dataset confidential — don't post either publicly or share them outside this process.

Good luck — we're more interested in your judgment than your polish.